

DIGT 3107 - Room/Hotel Booking System

Phase 3

Student#:	219941855	220329405	220486114
Sur (Family) Name:	Carter	Chawda	Dhawan
Given Name:	Toby	Gira	Komal
Section:	A	A	A

Query 1: Hotel's highest revenue week

This query calculates the revenue generated in each week of the hotel's operation, utilizing the Bookings, Rooms, and RoomTypes tables, and then accounting for staff discount on the rates, and returns the week with the highest revenue, and the total amount generated. This is useful for predicting future demand by being able to see which weeks the hotel generates the most revenue.

	week_start	week_end	total_revenue
▶	2024-12-22 14:00:00	2024-12-31 11:00:00	9845.000

Query 2: Check hotel double room type availability on a date

This query takes a date and returns all rooms that are available on that date, and also the selected check-in/check-out date the query is searching for, relying on the Rooms, RoomTypes, and Bookings tables. This is useful to get a quick picture of which rooms are available on any given night.

	room_number	selected_check_in	selected_check_out
▶	201	2025-03-02	2025-03-03
	202	2025-03-02	2025-03-03
	203	2025-03-02	2025-03-03
	204	2025-03-02	2025-03-03

Query 3: Total savings from all staff discounts year over year

This displays the total amount that has been saved by staff discounts in any given year. It utilizes the Bookings, Guests, Rooms, and RoomTypes tables to calculate this amount, and it is useful to see the impact of staff discounts on revenue, and the overall usage of the discount by the staff.

	year	total_savings
▶	2024	490.000
	2025	310.000

Query 4: Generate daily hotel check-in + check-out schedule + cancellations

This query is really a union of three queries, finding check-ins, check-outs, and cancellations for any given date, and ordering them by the time scheduled. This is incredibly helpful for organization on a daily basis for housekeeping, front-desk staff, and for managing any unexpected cancellations.

	guest_name	event_type	event_date
▶	Pepper Minstix	check_out	2024-12-25 10:00:00
	Frosty Snowman	cancellation	2024-12-25 14:00:00
	Santa Claus	check_in	2024-12-25 14:00:00
	Mrs. Claus	cancellation	2024-12-25 15:00:00
	Frosty Snowman	check_in	2024-12-25 16:00:00

Query 5: Frequent cancellation customer

This query finds the customer that has the most cancellations in the system using the Guests and Bookings tables, and displays the amount of bookings associated with them that have been cancelled. This is helpful for predicting future behavior, and potentially finding any fraudulent or dishonest behavior.

	guest_name	total_cancellations
▶	The Grinch	4

Query 6: Rooms that haven't been booked in the last month

This query calculates the rooms that have had no bookings made and associated with them within the past 30 days, relying on the booking_datetime of a Bookings table row and the Rooms table, and returns the last date the room was actually booked. This is a useful query to investigate which rooms are currently unpopular or underutilized.

NOTE: This query has an extra WHERE line ('AND booking_datetime < '2025-01-25') in it to allow us to assume for the purposes of this test that the current date is January 25th, 2025. In the real world, we'd usually use today's date, but for the purposes of testing with our mock data centered mostly on the Christmas season, we tested this query with the assumption that today is January 25th, 2025.

	room_number	max(b.booking_datetime)
▶	301	2024-12-01 09:30:00
	302	2024-12-02 10:15:00
	203	2024-12-24 14:30:00
	204	2024-12-24 15:45:00
	101	2024-12-20 09:00:00
	102	2024-12-21 10:30:00
	103	2024-12-21 11:15:00
	104	2024-12-22 08:45:00
	105	2024-12-22 09:30:00